SΔφ-26 - Irreversibility as Restoration Cost Explosion: Minimal Fixation Conditions after Pre-detection (v1.0)

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Core thesis: a transition becomes irreversible not when reversal is absolutely impossible, but when restoration cost exceeds the threshold of practical return.

This paper extends SΔφ-25 by clarifying when a latent or newly registered difference becomes a fixed event. The claim is minimal: irreversibility should not be treated as metaphysical finality or absolute undo impossibility. Instead, a transition becomes irreversible when return remains formally conceivable yet is no longer practically selectable within the operative system.

Under this view, eventhood depends on the economy of reversal. A difference becomes historically consequential when restoration cost rises beyond the system threshold for practical return. Thus, the world is not divided between absolutely reversible and absolutely irreversible transitions, but between low-cost return and cost-exploded fixation.

Keywords: irreversibility, reversibility, restoration cost, event fixation, rollback, threshold, pre-detection, SΔφ.

1. Problem

SΔφ-25 established that a difference may already be operative before detection. Yet not every operative difference becomes an event. Some differences dissipate, some remain latent, and some pass into registration without becoming historically decisive. The unresolved question is therefore not merely when a difference is detected, but when a difference becomes practically irreversible.

This paper proposes that irreversibility is best defined by restoration cost. What fixes an event is not the bare fact that something happened, but the closure of low-cost return.

2. Minimal definitions

Definition 1 - Reversibility
A state transition is reversible when restoration to a prior state remains available at sufficiently low cost.

Definition 2 - Irreversibility
A state transition is irreversible when restoration cost exceeds the threshold of practical return.

Definition 3 - Event fixation
An event becomes fixed when its reversal is no longer selected within the available cost structure.

Definition 4 - Restoration cost
Restoration cost is the total burden required to return from a present state to a prior state, including energetic, temporal, informational, networked, and interpretive costs.

3. Formal and practical reversibility

Formal reversibility means that prior state restoration remains logically describable or computationally representable. Practical irreversibility means that restoration cost rises beyond the threshold at which the system can or will no longer reverse the transition.

Formal reversibility does not guarantee practical reversibility. A system may still specify the prior state while lacking any economical path back to it. Therefore logical reversibility, computational reversibility, and actual return must be separated.

4. Minimal axioms

Axiom 1 - Irreversibility is economic before it is absolute

A transition is not irreversible because reversal is impossible, but because reversal is no longer economical within the operative system.

Axiom 2 - Event fixation follows cost escalation

Event fixation occurs when restoration cost rises faster than the system capacity or willingness to reverse.

Axiom 3 - Historical consequence begins at practical non-return

Practical irreversibility is the condition under which a transition becomes historically consequential.

Axiom 4 - Eventhood depends on closure of low-cost return

The eventhood of a transition depends not merely on its occurrence, but on the closure of low-cost return.

5. Minimal formulae

Restoration cost	C_restore(t) = f(E, T, I, N, M)
Reversibility condition	Reversible iff C_restore(t) < theta
Irreversibility condition	Irreversible iff C_restore(t) >= theta
Event fixation	Fixation(e) iff C_restore(e) >= theta

Here E denotes energetic or material cost, T temporal cost, I informational reconstruction cost, N network propagation cost, and M memory or interpretive residue cost. Theta denotes the system-dependent threshold of practical return.

6. Minimal examples

Broken cup. The prior configuration may remain physically describable, yet the cost required to reconstruct the exact prior state is prohibitively high. The transition is therefore practically irreversible.

Internet trace. A post may be deleted, but copies, screenshots, indexing, social propagation, and interpretive memory remain. Formal deletion does not restore the prior world-state at acceptable cost.

Virtual rollback. A simulated environment may allow rewind, but if observers, logs, external systems, or branching consequences retain the trace, rollback of one layer does not restore the total prior condition.

7. Consequence for event theory

Under this framework, many transitions occur, but only some become fixed events. A transition becomes an event in the full SΔφ sense only when a difference passes from latent activity into registration and restoration cost rises beyond the threshold of practical return.

Eventhood = detected difference + irreversible fixation.

8. Digital and virtual environments

Digital systems often appear reversible because copying is cheap, rollback is technically possible, and records can be edited or deleted. Yet digital environments may intensify irreversibility because replication is rapid, distributed observation multiplies traces, and restoration must address not one state but a network of propagated consequences.

Virtuality therefore does not necessarily weaken irreversibility. It may accelerate practical irreversibility by lowering propagation cost while raising restoration cost.

9. Final thesis

Irreversibility is restoration cost explosion. An event is fixed when return is no longer practically selectable. What fixes the world is not the impossibility of undoing, but the closure of acceptable return.

10. Series relation

SΔφ-25 explained how difference may exist before detection. SΔφ-26 explains when such difference becomes irreversibly fixed as an event. Together they define a minimal sequence: latent difference, registration threshold, restoration cost escalation, and event fixation.